Day 23 (Thu Nov 12): Files, modules, chart, conditional formatting, validation

Smooth | Python and Tools | covers only what was really done

Key ideas

- FileNotFoundError: the file is not there (a child of OSError). Relative paths depend on the folder Python runs from.
- with closes the file even if an error occurs inside; the error then continues out of the block.
- f.write adds no newline. Mode r reads (default), w erases on open, a adds at the end. open(file) with no mode is read-only, so writing raises UnsupportedOperation.
- Every line read from a file ends with "\n" (except possibly the last). strip() removes whitespace at both ends; split(",") cuts at commas; the result of strip() must be stored.
- The header row is data: int("marks") raises ValueError. next(f) reads and discards one line so the loop starts at line 2.
- math.sqrt always returns a float; round(7.5) = 8, round(6.5) = 6 (half to even); randint includes both ends; date minus date is a timedelta.
- Tools: column chart sorted high to low, axis from 0; Bottom 10% is a rank rule over the selected cells; a validation list from a range lives in one place.

Examples (run and verified)

```
"A,80\n".strip().split(",")   -> ['A', '80']
readlines()  -> ['Line 1\n', 'Line 2\n', ..., 'Line 5\n']      (5 items)
f.write("one"); f.write("two")  -> file holds 'onetwo'  (one line)

total = 0
with open("marks.csv") as f:
    next(f)
    for line in f:
        total += int(line.strip().split(",")[1])
# marks.csv: name,marks / A,80 / B,55  -> total 135
# header-only file -> total 0;  empty file -> next(f) raises StopIteration

math.sqrt(16) -> 4.0     math.floor(-7.1) -> -8      int(-7.1) -> -7
date(2026,12,31) - date(2026,12,25) -> 6 days, 0:00:00 ;  .days -> 6 (int)
days left from Nov 12 to Dec 31, 2026 -> 49
```

Interview edge

- A file object remembers its position, so next(f) and the for loop share it. Header-only gives 0, not an error; silent zeros are misleading.
- date.today() in Colab is usually UTC, so after midnight IST it can give yesterday. Days left excludes today; counting both ends gives 50.
- Excel ROUND rounds .5 away from zero; Python round() rounds half to even.

My mistakes and the fix

- Indentation of 5, 6 or 7 spaces inside a block raises IndentationError before any line runs. Fix: same spaces for every line in a block, 4 per level.
- Left quotes off the file name, called line.strip without brackets, and ignored the result of strip(). Fix: read the error name and message.
- Said StopIteration for a header-only file. It only happens for a file with zero lines.
- Wrote randint(1, 6) endpoints as 4 and 5, carrying the range rule into randint.
- Said a second write overwrites the first. Only opening with w erases.
- Answered a different question (last line without \n vs a line in the middle).

How I improved

- grade_of, the write-5-lines code and the days-left code were right with correct reasoning. The days-left prediction was written before running.
- Chose a bar chart and gave a non-circular reason; labelled blank regions "Unknown" instead of dropping them.
- Understood with, next(f) and Bottom 10% after correction.

Practice to retry (no solutions)

- Write 5 lines to a file with a loop, then read them back and print the count. Predict first.
- Print only the names from a name,marks CSV, then total the marks. Which row raises and which error?
- What do randint(0, 3) and range(0, 3) each give, and how many values?
- What does date(2026,12,25) minus date(2026,12,31) give? What does date + date raise?
- In Excel: bar chart of revenue by region sorted high to low; Bottom 10% in red; region drop-down from a range.

Recall questions (from memory)

- What do w, a and r do, and what is at the end of each line you read?
- What does next(f) do? What does a header-only file give versus an empty file?
- Does randint include its end? What does date minus date return?
- Why a bar chart for regions? Why is a range better than a typed list for a drop-down?

Open items

- Descriptive statistics, Text and date functions, Marks analyzer, NumPy arrays.
- Git conflict redo from a blank folder; README and .gitignore.